WORKED SOLUTION

STEP II 2013, Question 12

Probability and Statistics • Poisson Distribution Moments

STEP II 2013, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q12

The key idea

X keeps the odd values of U and Y keeps the even ones, so every expectation is a Poisson sum taken over only the odd or only the even terms. The trick for $\mathrm{E}(U)$ and $\mathrm{E}(U(U-1))$ still works: cancel k or $k(k-1)$ against $k!$ and pull out λ or λ^2. Each step down in k switches between odd and even, which is how α and β swap places. Note also that $\alpha+\beta=\mathrm{e}^{\lambda}$, so the factor $\mathrm{e}^{-\lambda}$ is $1/(\alpha+\beta)$.

Solution

Throughout, $\mathrm{P}(U=k)=\mathrm{e}^{-\lambda}\lambda^k/k!$ for $k=0,1,2,\ldots$, with $\lambda>0$. Adding the two series term by term gives

$$\alpha+\beta=\sum_{k\geq 0}\frac{\lambda^k}{k!}=\mathrm{e}^{\lambda},\qquad\text{so}\qquad \mathrm{e}^{-\lambda}=\frac{1}{\alpha+\beta}.$$

(In fact $\alpha=\cosh\lambda$ and $\beta=\sinh\lambda$.)

Part (i)

X is 0 unless U is odd, so only odd k contribute:

$$\mathrm{E}(X)=\sum_{k\ \text{odd}}k\,\frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}=\mathrm{e}^{-\lambda}\lambda\sum_{k\ \text{odd}}\frac{\lambda^{k-1}}{(k-1)!}=\mathrm{e}^{-\lambda}\lambda\left(1+\frac{\lambda^2}{2!}+\frac{\lambda^4}{4!}+\cdots\right),$$

because $k-1$ runs over $0,2,4,\ldots$. So

$$\mathrm{E}(X)=\frac{\lambda\alpha}{\alpha+\beta}.$$

In the same way, Y picks up the even $k\geq 2$ (the $k=0$ term is zero anyway), and then $k-1$ runs over $1,3,5,\ldots$:

$$\mathrm{E}(Y)=\mathrm{e}^{-\lambda}\lambda\sum_{k\ \text{even},k\geq 2}\frac{\lambda^{k-1}}{(k-1)!}=\frac{\lambda\beta}{\alpha+\beta}.$$

As a check, $\mathrm{E}(X)+\mathrm{E}(Y)=\lambda=\mathrm{E}(U)$, as it must be, because $X+Y=U$.

Part (ii)

Ask yourself How do we sum $k^2\lambda^k/k!$ when k^2 doesn't cancel neatly against $k!$?

Write $k^2=k(k-1)+k$. The $k(k-1)$ part cancels two factors of $k!$ and the k part cancels one. For odd k, the $k(k-1)$ sum starts at $k=3$ (it vanishes at $k=1$), and $k-2$ runs over $1,3,5,\ldots$:

$$\sum_{k\text{ odd}} k(k-1)\frac{\lambda^k}{k!}=\lambda^2\sum_{k\text{ odd},k\geq 3}\frac{\lambda^{k-2}}{(k-2)!}=\lambda^2\beta,\qquad \sum_{k\text{ odd}} k\frac{\lambda^k}{k!}=\lambda\alpha$$

(the second sum is the one from part (i)). Hence

$$\mathrm{E}(X^2)=\mathrm{e}^{-\lambda}\left(\lambda\alpha+\lambda^2\beta\right)=\frac{\lambda\alpha+\lambda^2\beta}{\alpha+\beta},$$

and subtracting $\mathrm{E}(X)^2$ gives

$$\mathrm{Var}(X)=\frac{\lambda\alpha+\lambda^2\beta}{\alpha+\beta}-\frac{\lambda^2\alpha^2}{(\alpha+\beta)^2},$$

as required.

For Y the parities swap. For even k the $k(k-1)$ sum starts at $k=2$ with $k-2$ running over $0,2,4,\ldots$, which gives $\lambda^2\alpha$. The k sum gives $\lambda\beta$, as in part (i). So

$$\mathrm{Var}(Y)=\frac{\lambda\beta+\lambda^2\alpha}{\alpha+\beta}-\frac{\lambda^2\beta^2}{(\alpha+\beta)^2}.$$

The final question. Because $X+Y=U$, we have $\mathrm{Var}(X+Y)=\mathrm{Var}(U)=\lambda$. Adding the two variances,

$$\begin{aligned}\mathrm{Var}(X)+\mathrm{Var}(Y)&=\frac{\lambda(\alpha+\beta)+\lambda^2(\alpha+\beta)}{\alpha+\beta}-\frac{\lambda^2(\alpha^2+\beta^2)}{(\alpha+\beta)^2}\\&=\lambda+\lambda^2\frac{(\alpha+\beta)^2-\alpha^2-\beta^2}{(\alpha+\beta)^2}=\lambda+\frac{2\lambda^2\alpha\beta}{(\alpha+\beta)^2}.\end{aligned}$$

Ask yourself Why should we expect the two sides to differ?

In general $\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$. Here $XY=0$ always, since at most one of X, Y is non-zero. So $\mathrm{Cov}(X,Y)=-\mathrm{E}(X)\mathrm{E}(Y)=-\lambda^2\alpha\beta/(\alpha+\beta)^2$, which matches the calculation above. Equality would need $\lambda^2\alpha\beta=0$. For $\lambda>0$, though, $\alpha\geq 1$ and $\beta\geq\lambda>0$, so $\lambda^2\alpha\beta>0$. Therefore **there are no non-zero values of** λ for which $\mathrm{Var}(X)+\mathrm{Var}(Y)=\mathrm{Var}(X+Y)$: the left-hand side is always strictly larger. In fact $\alpha=\cosh\lambda$ and $\beta=\sinh\lambda$, so the excess is $\lambda^2\sinh 2\lambda/\mathrm{e}^{2\lambda}=\frac{1}{2}\lambda^2\left(1-\mathrm{e}^{-4\lambda}\right)$.

Checks

- As $\lambda \to 0$, U is almost always 0 or 1, so Y is almost always 0. Indeed $\mathrm{E}(Y) \approx \lambda^2 \to 0$ and $\mathrm{E}(X) \approx \lambda$.
- For large λ, $\alpha \approx \beta \approx \frac{1}{2}\mathrm{e}^{\lambda}$, so $\mathrm{E}(X) \approx \mathrm{E}(Y) \approx \frac{1}{2}\lambda$: odd and even values become equally likely.
- At $\lambda = 1$, summing the Poisson probabilities directly gives $\mathrm{Var}(X) \approx 0.6778$ and $\mathrm{Var}(Y) \approx 0.8131$, so their sum exceeds $\mathrm{Var}(U) = 1$ by about $0.4908 = \frac{1}{2}(1 - \mathrm{e}^{-4})$. Both formulas agree.

Common slips

- Getting the parity wrong after cancelling: if k is odd, then $k-1$ is even (giving α) and $k-2$ is odd (giving β).
- Forgetting that $\mathrm{e}^{-\lambda} = 1/(\alpha+\beta)$, which leaves answers that aren't in the requested form.
- Assuming the variances add. X and Y are far from independent: knowing $X \neq 0$ forces $Y = 0$.

Beyond STEP

Splitting a series into its even and odd parts is the decomposition $\mathrm{e}^{\lambda} = \cosh\lambda + \sinh\lambda$. With probability generating functions it becomes routine: the even part of $G(s)$ is $\frac{1}{2}\big(G(s) + G(-s)\big)$. The covariance identity used at the end is one of the first tools in a university probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both generating functions and covariance.

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